Beyond the Invariant Measure: Ornstein's $\bar{d}$-Distance as a Dynamical-Fidelity Metric for Surrogate Models of Chaotic Systems

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Abstract

Learned surrogates of chaotic systems cannot be judged by pointwise trajectory error, because nearby trajectories separate exponentially. The de facto replacement is to compare *invariant measures* — where the system spends its time — via Wasserstein-type distances. But the invariant measure is a static object: two systems can share it exactly while disagreeing about every dynamical property (mixing, entropy production, temporal ordering). We import *Ornstein's $\bar{d}$-distance* from ergodic theory — a metric on stationary processes that metrizes dynamical, not merely distributional, equivalence — and turn it into a practical evaluation metric for chaotic surrogates. Our estimator computes exact optimal transport between empirical n-block distributions of symbolized trajectories under Hamming cost; every computed value is a statistically certified *lower* bound on $\bar{d}$, is reported against a same-process noise floor, and is complemented by an independent entropy-rate certificate obtained from a Fano-type inequality. On Lorenz-63 and the Kuramoto–Sivashinsky equation, surrogates constructed to match the invariant measure — IAAFT surrogates (exact marginal, matched spectrum) and time-rescaled flows (*exactly* the same invariant measure) — are invisible to measure- and spectrum-based baselines yet are separated from truth by $\bar{d}$ at more than $20\times$ the noise floor over 8 seeds (and remain detectable with as few as 10^4 samples). On a family of trained echo-state-network surrogates, $\bar{d}$ resolves dynamical failures that state-space Wasserstein cannot. We also report what the metric does *not* see (time reversal, at binary partitions), and release code.

1 Introduction

Data-driven surrogates of chaotic dynamics — neural operators, neural ODEs, reservoir computers — are now standard tools for emulating turbulence, climate, and other nonlinear systems. Evaluating them is not standard at all. Because chaotic systems have positive Lyapunov exponents, any pointwise comparison of trajectories becomes meaningless after a few Lyapunov times: a surrogate can be nearly perfect and still have large mean-squared error, and a surrogate frozen at the attractor's mean can beat it. The community's response has been to evaluate (and increasingly to train) surrogates against *long-run statistics*: compare the surrogate's invariant measure to the truth's, typically with Wasserstein or kernel distances on attractor samples [6, 11, 18, 25].

This paper starts from a structural observation: **the invariant measure is not the dynamics**. It records *where* the state spends time, not *how* it moves. The point is not a technicality

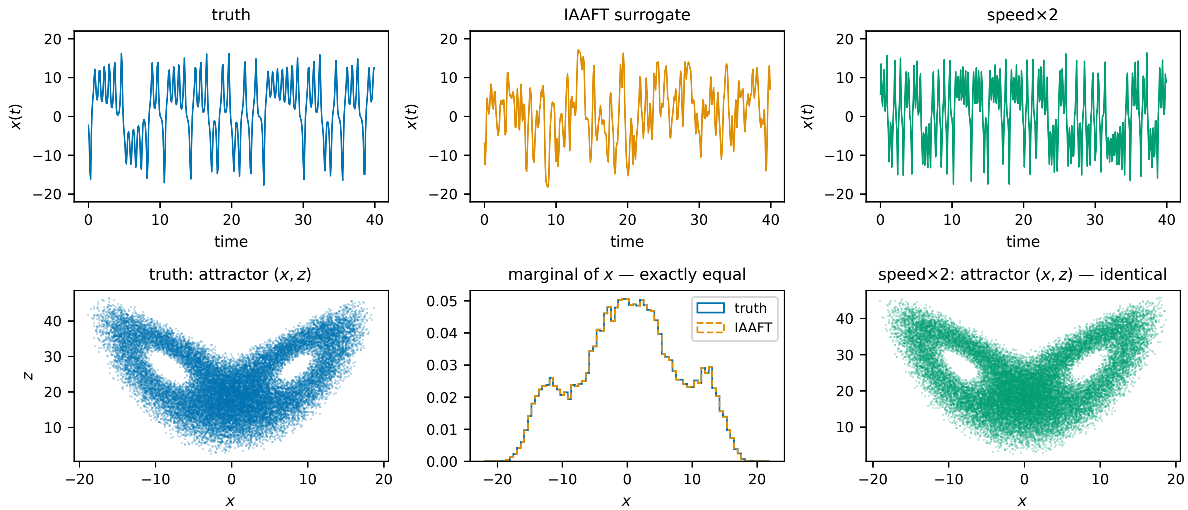

Figure 1: The failure mode this paper measures. *Top:* segments of the Lorenz-63 x coordinate for the true system, its IAAFT surrogate (identical marginal, matched spectrum), and the time-rescaled system $\dot{x} = 2f(x)$ (identical attractor). *Bottom:* the corresponding state-space densities are indistinguishable where defined — invariant-measure metrics certify all three as equivalent — yet the three processes are dynamically distinct, and $\bar{d}$ separates them (Table 1).

(Fig. 1). Run Lorenz-63 at double speed ($\dot{x} = 2f(x)$): the attractor, and with it the invariant measure, is *exactly* unchanged, while every temporal property — decorrelation, mixing rate, entropy production per unit time — changes by a factor of two. Apply an amplitude-adjusted Fourier-transform surrogate [26] to a Lorenz coordinate: the marginal distribution is preserved *exactly* and the power spectrum almost exactly, while the higher-order temporal structure that makes the signal deterministic chaos rather than colored noise is destroyed. A metric on invariant measures certifies both as perfect.

Ergodic theory has owned the right abstraction for half a century. Ornstein's $\bar{d}$-distance [8, 19] is a metric on *stationary processes* — laws of trajectories, not snapshots — under which two processes are close when their entire joint temporal structure can be aligned by a stationary coupling with small per-symbol disagreement. It is the metric of the Ornstein isomorphism theory: within the Bernoulli class, entropy is a complete invariant, and $\bar{d}$-limits preserve the dynamical properties (mixing, entropy) that static distributions do not determine. To our knowledge it has never been used to evaluate learned surrogates of chaotic systems.

Contributions.

1. **A practical estimator** (§3). We symbolize both systems with a common partition and compute exact optimal transport between empirical n-block distributions under normalized Hamming cost, for doubling block lengths n. Theory guarantees these values increase to $\bar{d}$ (Prop. 2.1), so each is a certified lower bound; a same-process noise floor computed with the identical budget makes finite-sample bias visible rather than deniable.

2. **An independent certificate** (§2.1). A Fano-type inequality bounds $\bar{d}$ from below by an explicit function of the symbolic block-entropy gap — a one-hour pre-check that needs no

transport computation. We state and use the finite-n form, which requires no entropy-rate limits.

3. **Validation on solvable processes** (§4): the estimator reproduces exact $\bar{d}_n$ values to four decimals, including a minimal pair of processes with identical marginals and maximally different dynamics.
4. **Decisive experiments** (§5–§6). On Lorenz-63 and Kuramoto–Sivashinsky, measure-matched adversarial surrogates are invisible to invariant-measure Wasserstein, spectrum, autocorrelation, and (for IAAFT) data-driven Lyapunov baselines, while $\bar{d}$ separates them from truth by more than an order of magnitude above the noise floor — stably across 8 seeds, three partitions, three sampling rates, and sample sizes down to 10^4.
5. **Learned surrogates** (§7): across an echo-state-network family, $\bar{d}$ resolves dynamical-fidelity differences among models that measure-based metrics rank as equally good, and agrees with all metrics when the surrogate is genuinely faithful.
6. **Honest negative results** (§9): time reversal — which preserves measure, spectrum, *and* entropy — is invisible to our estimator at binary partitions and reachable block lengths; we explain why (a structural symmetry of binary 2-blocks) and delimit what $\bar{d}$ adds relative to delay-embedded transport distances.

2 Background: processes, joinings, and $\bar{d}$

Let $X = (X_i)_{i\in\mathbb{Z}}$ and $Y = (Y_i)$ be stationary ergodic processes on a finite alphabet $\mathcal{A}$, with laws μ, ν on $\mathcal{A}^{\mathbb{Z}}$. In our setting these arise by sampling a flow at interval τ and applying a partition $\pi : \mathbb{R}^d \to \mathcal{A}$ to the state (§3); the process law encodes the full temporal structure of the symbolized dynamics.

Definition 2.1 (Joining; $\bar{d}$). *A* joining λ *of* (μ, ν) *is a shift-invariant probability measure on* $(\mathcal{A}\times\mathcal{A})^{\mathbb{Z}}$ *with marginals* μ *and* ν*. Ornstein's distance is*

$$\bar{d}(\mu,\nu) \;=\; \inf_{\lambda \text{ joining}} \lambda(x_0 \neq y_0)\,.$$

Equivalently, $\bar{d}$ is the Wasserstein-1 distance between process laws under the normalized Hamming metric on sequences [8]: the minimal asymptotic fraction of time at which the two processes must disagree, over all ways of running them jointly while preserving each one's statistics. $\bar{d}(\mu,\nu) = 0$ iff $\mu = \nu$; two surrogates with the same invariant measure but different temporal laws are at positive $\bar{d}$.

For $n \geq 1$ let P_n, Q_n denote the n-block marginals of μ, ν on $\mathcal{A}^n$, and

$$\bar{d}_n(\mu,\nu) \;=\; W_1^{\text{Ham}}(P_n, Q_n) \;=\; \min_{\gamma\in\Gamma(P_n,Q_n)} \mathbb{E}_{(a,b)\sim\gamma}\Big[\tfrac{1}{n}\textstyle\sum_{i=1}^n \mathbf{1}\{a_i \neq b_i\}\Big],$$

a finite optimal-transport problem.

Proposition 2.1 (n-block characterization). *For stationary* μ, ν*: (i)* $\bar{d}_n(\mu,\nu) \leq \bar{d}(\mu,\nu)$ *for every* n*; (ii)* $n\,\bar{d}_n$ *is superadditive, hence* $\bar{d}_{2n} \geq \bar{d}_n$ *and (iii)* $\bar{d}_n \to \sup_m \bar{d}_m = \bar{d}$ *as* $n \to \infty$ *[8].*

Proofs of the finite-n statements are elementary and given in Appendix A. Three consequences shape the method. First, every computed $\bar{d}_n$ is a *certified lower bound* on $\bar{d}$: plateau values understate, never overstate, the distance. Second, along the doubling schedule $n = 1, 2, 4, \ldots$ the true values are monotone, so a rising-then-flat empirical curve is the theoretically expected signature. Third, $\bar{d}_1$ is exactly the total-variation distance between the 1-symbol marginals — i.e. the invariant measure comparison at partition resolution — so the decomposition of $\bar{d}_n$ over n makes "static vs. dynamical" quantitative: a measure-matched surrogate has $\bar{d}_1 \approx 0$, and everything it gets wrong appears at $n \geq 2$.

2.1 Entropy is $\bar{d}$-continuous: a free certificate

Write $H_n(\mu)$ for the Shannon entropy (bits) of P_n and $h(\mu) = \lim_n H_n(\mu)/n$ for the entropy rate. Entropy rate is continuous with respect to $\bar{d}$ (27, §I.9; see also 17), and the modulus is explicit:

Proposition 2.2 (Fano-type bound, finite-n form)**.** *Let $g(\delta) = H_{\mathrm{b}}(\delta) + \delta \log_2(|\mathcal{A}| - 1)$, with H_{b} the binary entropy function. For every n,*

$$\frac{|H_n(\mu) - H_n(\nu)|}{n} \;\leq\; g\big(\bar{d}_n(\mu,\nu)\big) \;\leq\; g\big(\bar{d}(\mu,\nu)\big), \qquad \textit{hence} \qquad \bar{d} \;\geq\; g^{-1}\!\left(\frac{|H_n(\mu) - H_n(\nu)|}{n}\right).$$

Block entropies are estimable by counting; the bound therefore converts an entropy gap into a certified lower bound on $\bar{d}$ *without solving any transport problem.* We use it in two roles: as a cheap go/no-go pre-check, and as an independent pipeline whose agreement with the transport estimator (estimate $\geq$ certificate) is a consistency test of both. We emphasize the finite-n form because it avoids estimating entropy-rate limits; the commonly quoted form $\bar{d} \geq |h(\mu) - h(\nu)|$ is *not* correct as stated — inverting g is necessary.[1]

2.2 What can and cannot be estimated

A subtlety deserves daylight. Ornstein and Weiss [20] proved that universal consistent estimation of a process in $\bar{d}$ from samples is possible on the class of B-processes and, in a precise sense, on no larger class. We therefore do *not* claim a universal consistent estimator of $\bar{d}$ itself. Our claims are weaker and safe: each $\bar{d}_n$ is a functional of the n-dimensional marginals and is consistently estimable for stationary ergodic data (the empirical n-block distribution converges by the ergodic theorem, and W_1^{Ham} is continuous on the finite simplex); by Prop. 2.1 the reported $\max_{n \in \mathcal{N}} \bar{d}_n$ is a lower bound on $\bar{d}$ whose tightness increases with n, with no uniform rate over process pairs. For an evaluation metric this is the right shape: detections are certified, and non-detections are explicitly inconclusive.

3 The estimator

Symbolization. Choose a partition $\pi : \mathbb{R}^d \to \mathcal{A}$ and a sampling interval τ; both systems are symbolized with the *same* π (fit on truth data where data-dependent, e.g. quantile edges) — the partition is part of the metric, and using different partitions per system would confound the comparison. Defaults in this paper: $\pi = \mathrm{sign}(x)$ (which lobe of the Lorenz attractor; median split for KS), $\tau = 0.1$ (Lorenz), 1.0 (KS). Sensitivity to both choices is quantified in §5.1.

[1] For binary alphabets $g = H_{\mathrm{b}}$, and $g^{-1}(x) \leq x$ with equality only at 0: e.g. an entropy gap of 1 bit certifies only $\bar{d} \geq \frac{1}{2}$, which is sharp (Appendix A).

Transport on n-blocks. From symbol streams of length N we form all overlapping n-blocks, encode them as integers, and deduplicate into weighted supports. When both supports have size ≤ 4096 we solve the exact transport problem between the full empirical distributions (network simplex, POT [5]); otherwise we draw a fixed budget of B random blocks per side (point-cloud OT, here $B \in \{2000, 4000\}$). The Hamming cost matrix is computed by XOR/popcount on bit-packed blocks. We evaluate $n \in \{1, 2, 4, 8, 16, 32\}$; a curve costs seconds to tens of seconds on one CPU core.

Noise floors, not error bars alone. The empirical OT distance between two independent samples of the *same* process is strictly positive at finite N (empirical measures never coincide), and this bias grows with n in the sampled regime. We therefore report, for every pair and every n, the same-process distances (truth-vs-truth and surrogate-vs-surrogate on disjoint data halves) computed with the identical regime and budget. Only separation above the floor counts as signal; the floor makes the estimator's finite-sample resolution an explicit, published quantity.

Entropy pipeline. Block entropies H_n with Miller–Madow correction and an undersampling guard (trust n only while observed support $\leq 0.02\times$ window count), conditional entropies $h_n = H_n - H_{n-1}$, and an independent LZ78 estimate [15, 31] as a cross-check; certificates via Prop. 2.2.

4 Validation on exactly solvable processes

Three families with known answers (details and proofs in Appendix A; figure 2):

1. **iid Bernoulli(p) vs iid Bernoulli(q):** $\bar{d}_n = |p - q|$ exactly, for every n. With $p, q = (0.3, 0.5)$, $N = 10^6$: full-regime estimates match to $< 5 \times 10^{-4}$; sampled-regime estimates match within their published floors.
2. **iid($\frac{1}{2}$) vs period-2 alternation** (random phase): identical 1-marginals, so $\bar{d}_1 = 0$ — a static metric sees nothing — while $\bar{d}_n = \mathbb{E}[\min(k, n-k)]/n$, $k \sim \mathrm{Bin}(n, \frac{1}{2})$, rising to $\bar{d} = \frac{1}{2}$ (Prop. A.1). The estimator reproduces 0.25, 0.3125, 0.3633, … to four decimals. The Fano certificate independently forces $\bar{d} \geq H_{\mathrm{b}}^{-1}(1\,\mathrm{bit}) = \frac{1}{2}$: both pipelines agree, and the example is the minimal instance of the failure mode this paper is about.
3. **Markov chains:** estimated $\bar{d}$ plateau $\geq$ Fano certificate holds (consistency), and entropy estimators hit $H_{\mathrm{b}}(0.1)$ within 0.007 bits.

5 Lorenz-63: constructed adversaries

Setup. Lorenz-63 [16] ($\sigma, \rho, \beta = 10, 28, 8/3$; RK4, $dt = 0.005$; sampled at $\tau = 0.1$; $N = 5 \times 10^5$ per run). All results: mean $\pm$ std over **8 independent realizations** (independent truth run *and* independently constructed surrogates per seed). Symbolic truth entropy: 0.2214 ± 0.0003 bits/symbol. The Rosenstein divergence-rate diagnostic reads $\lambda_1 = 1.189 \pm 0.020$ nats/unit time under our fixed protocol (literature $\lambda_1 = 0.906$ [4, 24]; the absolute value is fit-window dependent, Appendix B — we use it only comparatively, under one frozen protocol for all systems).

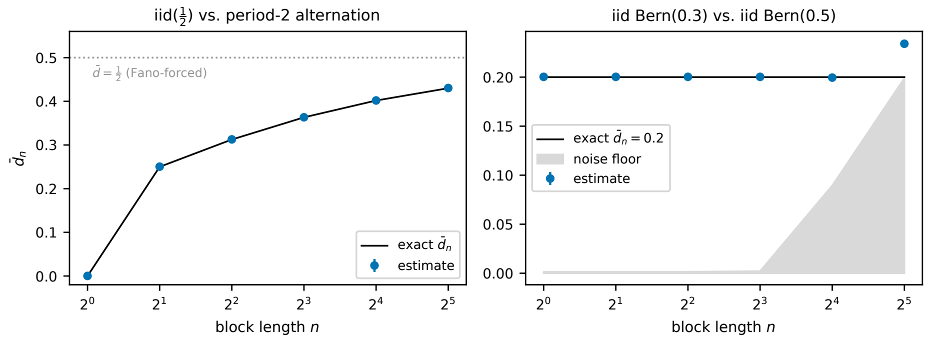

Figure 2: Estimator validation against closed forms ($N = 10^6$). *Left:* iid($\frac{1}{2}$) vs. period-2 alternation — identical marginals ($\bar{d}_1 = 0$), exact $\bar{d}_n = \mathbb{E}[\min(k, n-k)]/n$ (line) vs. estimates (points). *Right:* iid Bernoulli(0.3) vs. (0.5) — exact $\bar{d}_n = 0.2$ at every n; grey: same-process noise floor at the identical budget (the visible large-n inflation of the raw estimate is exactly what the floor protocol quantifies).

Surrogate zoo. Each construction defeats a specific baseline family *by design*, with zero training:

- **independent truth** — negative control; every metric should read zero/floor.
- **IAAFT** [26, 29] of $x(t)$: *exactly* the truth marginal, matched power spectrum, scrambled higher-order temporal structure. (Surrogate-data methods usually construct null models to test data for nonlinearity; we repurpose them as adversaries to test *metrics*.)
- **speed**$\times$2: integrate $\dot{x} = 2f(x)$ — *exactly* the truth attractor and invariant measure, twice the entropy production per unit time.
- **time-reversed**: identical measure, identical spectrum (ACF is even), identical entropy rate — hard mode: only genuinely temporal-order-sensitive metrics can see it.
- $\rho = 32$: a genuinely different attractor — positive control.

Results (Table 1, Fig. 3). The pattern the paper exists to demonstrate:

- **IAAFT defeats every standard baseline at once**: W_1 marginal $= 0.0052 \pm 0.0052$ (negative control: 0.0052 ± 0.0052), PSD distance $0.40 \pm 0.01\,\mathrm{dB}$ (control: 0.40 ± 0.01), ACF distance at control level; the Rosenstein λ_1 fit degenerates (low r^2) because there is no deterministic divergence to fit — data-driven Lyapunov estimation presupposes determinism. Meanwhile $\bar{d}$ separation is 0.1045 ± 0.0018, i.e. $\sim 28\times$ the noise floor, and the entropy certificate independently guarantees $\bar{d} \geq 0.0619 \pm 0.0005$.
- **speed$\times$2 is certified perfect by the invariant-measure metric** (W_1 state $= 0.128 \pm 0.010$, at the same-vs-same floor 0.130) while $\bar{d}$ reads 0.1074 ± 0.0020. The divergence-rate diagnostic rises sharply (the underlying λ_1 doubles by construction; the fixed-protocol Rosenstein slope ratio is ≈ 1.5, compressed by fit-window saturation — see Appendix B for the window study). The spectrum does flag this surrogate — which is exactly why both adversaries are needed: IAAFT kills the spectrum baseline, speed$\times$2 kills the measure baseline, and $\bar{d}$ catches both.

Table 1: Lorenz-63, mean $\pm$ std over 8 seeds. *Top:* static and linear baselines (Wasserstein distances normalized by attractor scale; parentheses: same-vs-same floor). *Bottom:* dynamical metrics (λ_1 in nats per unit time with linear-fit r^2; h = symbolic entropy, bits/symbol; Fano LB and $\bar{d}$ separation as defined in §3). Truth reference: $h = 0.2214 \pm 0.0003$, $\lambda_1 = 1.189 \pm 0.020$.

	W_1 marginal	W_1 state (3D)	W_1 delay	PSD (dB)	ACF
independent truth (neg. ctrl)	0.0052 ± 0.0052	0.129 ± 0.005 (0.130)	0.094 ± 0.006 (0.102)	0.40 ± 0.01	0.0040 ± 0.0006
IAAFT	0.0052 ± 0.0052	—	0.366 ± 0.006 (0.102)	0.40 ± 0.01	0.0040 ± 0.0006
speed×2	0.0067 ± 0.0039	0.128 ± 0.010 (0.130)	0.634 ± 0.007 (0.102)	10.56 ± 0.01	0.0408 ± 0.0005
time-reversed	0.0052 ± 0.0052	0.129 ± 0.012 (0.130)	0.223 ± 0.010 (0.102)	0.40 ± 0.01	0.0040 ± 0.0006
$\rho = 32$ (pos. ctrl)	0.0806 ± 0.0007	0.496 ± 0.016 (0.130)	0.185 ± 0.010 (0.102)	1.97 ± 0.01	0.0086 ± 0.0008

	λ_1 [r^2]	h (bits/sym)	Fano LB on $\bar{d}$	$\bar{d}$ (floor)	$\bar{d}$ sep
independent truth (neg. ctrl)	1.194 ± 0.020 [0.98]	0.2213 ± 0.0004	0.0001 ± 0.0000	0.0032 ± 0.0022 (0.0041)	-0.0008 ± 0.0027
IAAFT	0.778 ± 0.011 [0.65]	0.6340 ± 0.0016	0.0619 ± 0.0005	0.1082 ± 0.0006 (0.0037)	0.1045 ± 0.0018
speed×2	1.804 ± 0.019 [1.00]	0.3383 ± 0.0002	0.0195 ± 0.0002	0.1123 ± 0.0009 (0.0049)	0.1074 ± 0.0020
time-reversed	3.051 ± 0.026 [0.97]	0.2213 ± 0.0004	0.0001 ± 0.0000	0.0052 ± 0.0018 (0.0048)	0.0004 ± 0.0020
$\rho = 32$ (pos. ctrl)	1.177 ± 0.008 [0.98]	0.2251 ± 0.0002	0.0020 ± 0.0002	0.0423 ± 0.0012 (0.0053)	0.0369 ± 0.0017

Table 2: Floor-adjusted $\bar{d}$ separation under partition and sampling variations (Lorenz-63, single seed).

variation	neg. control	IAAFT	speed×2
sign(x), m=2, τ=0.1 (reference)	−0.0012	0.1057	0.1102
quantile-4(x), m=4	−0.0021	0.1660	0.3200
box(x, z), m=6	0.0004	—	0.3943
sign(x), τ=0.05	−0.0099	0.0896	0.1023
sign(x), τ=0.25	−0.0001	0.1149	0.0952

- **The positive control is caught by both families** (W_1 state 0.496 ± 0.016; $\bar{d}$ sep 0.0369 ± 0.0017) — notably $\bar{d}$ detects it although its entropy gap is negligible (0.0020 ± 0.0002): the transport estimator carries information beyond the entropy certificate.
- **The negative control is clean** on every metric, and $\bar{d}$'s floor protocol reads it as exactly "no signal" (-0.0008 ± 0.0027).
- **Time reversal is invisible to $\bar{d}$ at this partition** (0.0004 ± 0.0020) — see §9 for why, and for what does see it.

Sample efficiency. Repeating the IAAFT and speed×2 comparisons at $N = 10^4, 10^5, 10^6$ gives peak separations of 0.093/0.087/0.106 and 0.099/0.109/0.110 respectively: the estimate is stable two orders of magnitude below the data sizes typical of surrogate evaluation.

5.1 Partition and timescale sensitivity

The partition is the estimator's main free parameter (a known confound: generating partitions are unknown for general systems [3, 12]). Table 2 repeats the core comparison under three partitions and three sampling intervals (single realization; the multi-seed variation in Table 1 calibrates noise). Rankings never flip; refinement *increases* separation. Conclusions do not depend on the partition within the tested family.

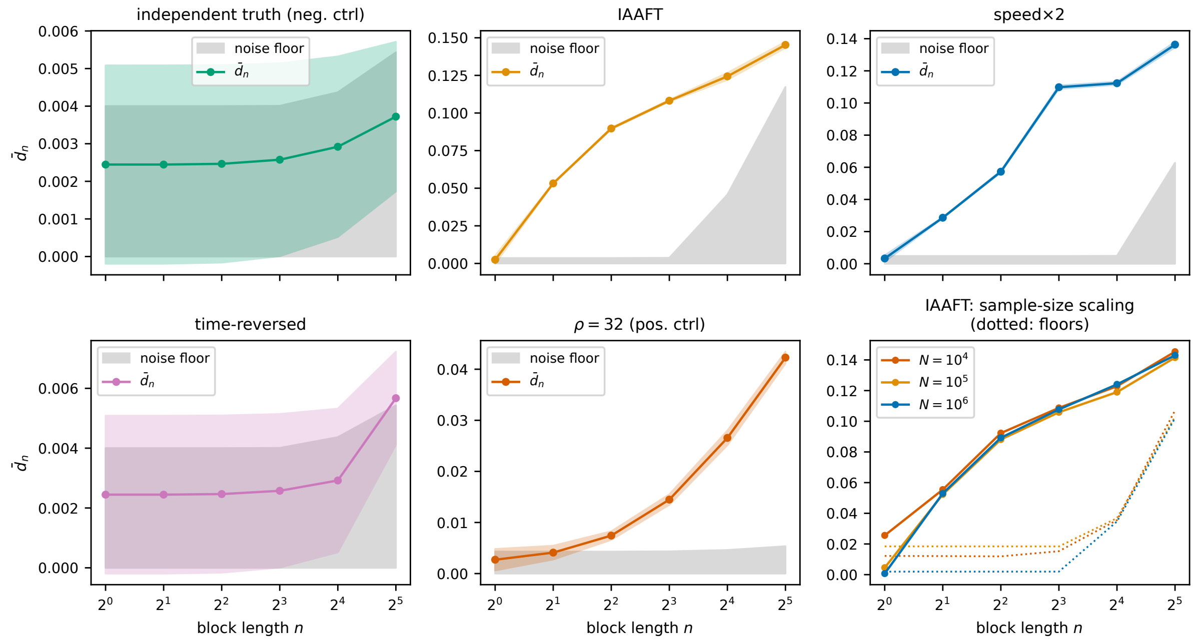

Figure 3: $\bar{d}_n$ convergence curves (mean over 8 seeds; bands: ± 1 std; grey: same-process noise floor at identical budget). Curves rise monotonically along doubling n and flatten, as Prop. 2.1 predicts. $\bar{d}_1 \approx 0$ for every measure-matched surrogate: all discrimination lives in temporal structure.

6 Kuramoto–Sivashinsky: spatiotemporal chaos

To test that nothing here is Lorenz-specific, we repeat the construction on the Kuramoto–Sivashinsky equation [14, 28], $u_t = -uu_x - u_{xx} - u_{xxxx}$ on $[0, L]$ periodic, at the standard chaotic size $L = 22$ [2] (ETDRK4 [13], $N = 64$ modes, $dt = 0.25$, sampled at $\tau = 1.0$; 2×10^5 samples; 4 seeds).

KS adds a methodological wrinkle worth recording: the equation has $O(2)$ symmetry and its spatial phase diffuses slowly, so *non*-invariant observables confound dynamics with phase position. Working with raw states and the pointwise observable $u(x_0, t)$, the same-process floors correctly flagged this — two *independent truth* runs showed a 64-dimensional state W_1 of ≈ 2.3 (attractor-scale units) purely from phase, and elevated, seed-dependent symbol floors from slow drift (archived in the repository). The remedy is standard symmetry reduction: we compare states through translation-invariant coordinates (leading Fourier magnitudes $|\hat{u}_1|, \ldots, |\hat{u}_{10}|$) and symbolize the scalar observable $|\hat{u}_1|(t)$ with a median split. For systems with continuous symmetries this should be considered part of the partition choice. Surrogates: independent truth, IAAFT of the observable, and the time-rescaled PDE $u_t = 2(\cdot)$ — again *exactly* the same invariant measure. Truth symbolic entropy: 0.3479 ± 0.0005 bits/symbol.

The pattern replicates (Table 3): the time-rescaled PDE is statistically indistinguishable from the same-vs-same floor on the symmetry-reduced state W_1 — indeed *below* the independent-truth control — while $\bar{d}$ separates it by 0.1420 ± 0.0010, $\approx 20\times$ its same-process floor (the negative control reads 0.0005); IAAFT matches the marginal and the spectrum of the observable to control level while $\bar{d}$ separates it (0.0395 ± 0.0016); the negative control is clean. Both detections

Table 3: Kuramoto–Sivashinsky ($L = 22$), mean $\pm$ std over 4 seeds. W_1 state is computed between point clouds of translation-invariant Fourier-magnitude coordinates (floor in parentheses); marginal, PSD and symbols use the observable $|\hat{u}_1|(t)$.

	W_1 marginal	W_1 state (sym. red.)	PSD (dB)	h (bits/sym)	Fano LB	$\bar{d}$ sep
independent truth (neg. ctrl)	0.0066 ± 0.0038	0.558 ± 0.025 (0.520)	0.64 ± 0.02	0.3482 ± 0.0008	0.0002 ± 0.0001	0.0005 ± 0.0025
IAAFT	0.0066 ± 0.0038	—	0.64 ± 0.02	0.5002 ± 0.0014	0.0020 ± 0.0001	0.0395 ± 0.0016
speed×2	0.0055 ± 0.0019	0.533 ± 0.009 (0.520)	12.79 ± 0.02	0.4364 ± 0.0017	0.0295 ± 0.0004	0.1420 ± 0.0010

peak at $n = 16$ in the exact-support regime. The metric transfers to spatiotemporal chaos without modification.

7 Learned surrogates: an echo-state-network family

Constructed adversaries prove discrimination; reviewers of surrogate models rightly ask about *trained* ones. We train echo state networks [10, 22] on Lorenz-63 (400-unit reservoir, $[r; r^2]$ readout features, ridge regression, 10^5 training samples at $\tau = 0.1$), sweep the reservoir spectral radius $\rho_s \in [0.6, 1.4]$ (two reservoir seeds each), and evaluate 3×10^5-step *autonomous* rollouts against an independent truth run.

Results (Table 4, Fig. 4) split into three regimes. *First, no false alarms:* across the healthy sweep, every rollout that reproduces the invariant measure (W_1 at its same-vs-same floor) also gets the dynamics right, and $\bar{d}$ agrees — its separation stays within ± 0.008 of zero, i.e. within floor noise. A dynamical metric that fired on healthy surrogates would be useless; this one does not. *Second, gross failures are caught by everything:* one $\rho_s = 1.4$ rollout leaves the attractor for a low-entropy orbit ($W_1 = 1.84$, $\bar{d}$ sep $= 0.39$, symbolic entropy 0.020 vs. truth 0.221), and the tiny reservoir either diverges or collapses; measure- and process-level metrics agree. *Third — the regime that matters — subtle dynamical corruption dissociates the metrics:* removing the $[r; r^2]$ readout's quadratic half (the standard device for handling the Lorenz symmetry [22]) yields rollouts with excellent one-step error ($\sim 2 \times 10^{-3}$) whose invariant-measure distance is marginal — $W_1 = 0.152$ against a floor of 0.127 for one reservoir seed — while $\bar{d}$ separates both seeds (0.019 and 0.033, i.e. 2–4× their $\bar{d}$ floor of 0.008), clearly outside the healthy band in which every model sat within $\pm$one floor of zero, and the entropy certificate moves in agreement ($h = 0.230, 0.260$ vs. truth 0.221). The undertrained and over-regularized controls remain faithful on both axes — degradation per se is not what $\bar{d}$ detects; *dynamical* degradation is.

8 Related work

Invariant-measure training and evaluation. DySLIM [25] regularizes training toward the invariant measure; Jiang et al. [11] match invariant measures of neural operators with optimal-transport and contrastive losses; Melo et al. [18] use adversarial OT objectives on attractor statistics; Fumagalli et al. [6] build evaluation measures from correlation integrals and attractor densities. All compare static objects; the present work is complementary — it metrizes the process.

Dynamical diagnostics. Lyapunov spectra [4, 24], entropy production [7, 23], and spectra/ACF are classical per-system invariants; matching them is necessary but not sufficient for

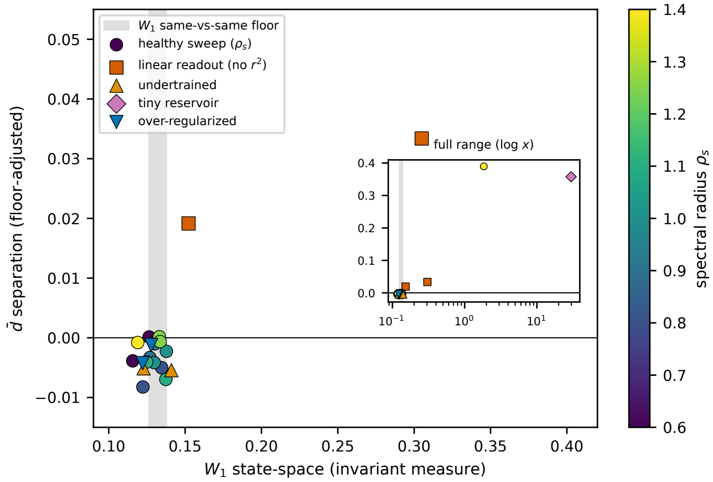

Figure 4: ESN family: invariant-measure fidelity (W_1 state, x-axis; grey band: same-vs-same floor) vs. dynamical fidelity ($\bar{d}$ separation, y-axis), colored by spectral radius. Healthy models cluster at the origin (measure-faithful *and* dynamics-faithful); gross failures leave along both axes; the linear-readout models (red squares) separate vertically: wrong dynamics at near-floor invariant-measure distance.

process equivalence, and estimating them from surrogate *output* presumes determinism (our IAAFT results make that concrete). Herz et al. [9] document, in the probabilistic-forecasting setting, the same tension between finite-horizon objectives and dynamical invariants that motivates us; their response is a training framework, ours an evaluation metric with certificates. Long-run climate statistics of reservoir computers are studied in [22, 30]. A parallel motivation appears in neuroscience: Dynamical Similarity Analysis [21] compares two systems by the temporal structure of their computation — via a Koopman/Procrustes alignment of learned linear embeddings — rather than the geometry of their states, drawing the same distributional-versus-dynamical distinction we do, in a different domain and with different machinery ($\bar{d}$ is a metric on symbolic processes with certified bounds, not an embedding alignment).

Ergodic-theoretic tools. $\bar{d}$ originates in the isomorphism theory of Bernoulli shifts [19]; Gray et al. [8] generalized it and connected it to information theory; entropy's $\bar{d}$-continuity and the block characterizations are classical [27]. Estimability limits are due to Ornstein and Weiss [20]. Symbolic dynamics as a data-analysis device is reviewed in [3]. Our contribution is the transfer of this machinery to surrogate evaluation, with a floor-calibrated, certificate-carrying estimator.

9 Limitations and outlook

What the metric does not see. Time reversal preserves the invariant measure, the power spectrum, and the entropy rate; at a binary partition it also preserves all 2-block statistics *exactly* (for any stationary binary process, $P(01) = P(10)$ by transition counting), so detection must come from $n \geq 3$ asymmetries, which are evidently below our floors here. The delay-embedded W_1 — itself a temporal object, not an invariant-measure metric — does detect reversal and IAAFT, and is the strongest baseline in our matrix; it lacks, however, $\bar{d}$'s block-length

Table 4: ESN surrogates of Lorenz-63 (each row: mean over 2 reservoir seeds). One-step NRMSE is teacher-forced on held-out data; remaining metrics evaluate the autonomous rollout. *div/col* = diverged or collapsed before 5×10^4 steps.

model	1-step NRMSE	status	W_1 state (floor)	h (bits/sym)	Fano LB	$\bar{d}$ sep
$\rho_s = 0.60$	1.4×10^{-5}	ok	0.121 (0.132)	0.2218	0.0001	−0.0019
$\rho_s = 0.80$	1.1×10^{-4}	ok	0.129 (0.132)	0.2216	0.0000	−0.0066
$\rho_s = 0.90$	2.9×10^{-4}	ok	0.129 (0.132)	0.2217	0.0000	−0.0022
$\rho_s = 1.00$	7.1×10^{-4}	ok	0.134 (0.132)	0.2221	0.0002	−0.0032
$\rho_s = 1.10$	1.6×10^{-3}	ok	0.131 (0.132)	0.2214	0.0001	−0.0055
$\rho_s = 1.25$	4.9×10^{-3}	ok	0.133 (0.132)	0.2220	0.0001	−0.0002
$\rho_s = 1.40$ (seed 0)	1.2×10^{-2}	ok	0.119 (0.127)	0.2244	0.0004	−0.0008
$\rho_s = 1.40$ (seed 1)	1.5×10^{-2}	ok	1.839 (0.138)	0.0197	0.2323	0.3898
linear readout (no r^2)	1.8×10^{-3}	ok	0.229 (0.132)	0.2449	0.0020	0.0263
undertrained (2×10^3 samples)	2.6×10^{-3}	ok	0.132 (0.132)	0.2225	0.0002	−0.0053
tiny reservoir ($n_r = 50$)	5.1×10^{-3}	diverged/ok	29.713 (0.138)	0.0258	0.1970	0.3576
over-regularized (ridge 10^{-1})	7.7×10^{-4}	ok	0.125 (0.132)	0.2218	0.0001	−0.0027

decomposition, certified lower bounds, and clean null behavior. A finer partition and/or larger n (with the attendant sample cost) is the principled route to reversibility within $\bar{d}$.

Partition choice. All statements are relative to the chosen partition: $\bar{d}$ on symbols is a lower bound on dynamical discrepancy, and a coarse partition can hide real differences (reversal, above). Our sensitivity protocol (fit partition on truth, apply to both, vary the family, report all) is a mitigation, not a solution; partition learning is an obvious next step.

Estimation. Universal rates are impossible (§2.2); floors grow with n in the sampled regime, capping usable block lengths at fixed budgets. All our detections occur already in the exact-support regime (at $n \leq 16$ for every measure-matched adversary; $n = 32$ for the ρ-perturbed control).

Outlook. The Sinkhorn relaxation of the n-block transport problem is differentiable in the underlying trajectory through soft symbolization [1], making $\bar{d}$ a candidate *training* signal — a DySLIM-style regularizer that targets the process rather than the measure. Our results suggest that is the right object to optimize; building it is future work.

Reproducibility. All experiments run on CPU in minutes-to-tens-of-minutes. Code, seeds, raw results, and the exact scripts that generate every number and figure in this paper are archived at https://doi.org/10.5281/zenodo.21168867.

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A Proofs

A.1 Proposition 2.1

(i) A joining λ with $\lambda(x_0 \neq y_0) = \bar{d}$ has, by shift invariance, per-coordinate disagreement $\bar{d}$ at every coordinate; its restriction to coordinates $1..n$ is a coupling of (P_n, Q_n) with normalized Hamming cost $\bar{d}$. Hence $\bar{d}_n \leq \bar{d}$. *(ii)* Let γ be an optimal coupling for $\bar{d}_{n+m}$ and let δ_i be its disagreement probability at coordinate i. Restrictions to coordinates $1..n$ and $n+1..n+m$ are couplings of (P_n, Q_n) and (P_m, Q_m) (stationarity), so $(n+m)\,\bar{d}_{n+m} = \sum_{i \leq n} \delta_i + \sum_{i > n} \delta_i \geq n\,\bar{d}_n + m\,\bar{d}_m$. Superadditivity gives $\bar{d}_{2n} \geq \bar{d}_n$ and, by Fekete's lemma, $\lim_n \bar{d}_n = \sup_n \bar{d}_n \leq \bar{d}$. *(iii)* Equality of the limit with $\bar{d}$ is the process-level identity of Gray et al. [8]: optimal n-block couplings have subsequential weak limits that are joinings (a diagonal/averaging argument restores shift invariance), with cost $\lim_n \bar{d}_n$. □

A.2 Proposition 2.2

Let γ be an optimal coupling of (P_n, Q_n) with per-coordinate disagreements δ_i, $\frac{1}{n}\sum \delta_i = \bar{d}_n$. Then $H(X^n) \leq H(Y^n) + H(X^n \mid Y^n) \leq H(Y^n) + \sum_i H(X_i \mid Y^n) \leq H(Y^n) + \sum_i H(X_i \mid Y_i)$, and Fano's inequality gives $H(X_i \mid Y_i) \leq H_{\mathrm{b}}(\delta_i) + \delta_i \log_2(|\mathcal{A}| - 1)$. Concavity of g and Jensen yield $\frac{1}{n}\sum_i g(\delta_i) \leq g(\bar{d}_n)$. Symmetrize in (μ, ν) and use monotonicity of g on $[0, 1 - 1/|\mathcal{A}|]$ with $\bar{d}_n \leq \bar{d}$. □

A.3 iid Bernoulli pairs

For product measures with per-coordinate marginals p, q: any coupling of (P_n, Q_n) has $P(x_i \neq y_i) \geq \mathrm{TV}(\mathrm{Bern}(p), \mathrm{Bern}(q)) = |p - q|$ at each i, so $\bar{d}_n \geq |p - q|$; the product of per-coordinate monotone couplings achieves it. Hence $\bar{d}_n = |p - q|$ for all n, and $\bar{d} = |p - q|$. □

A.4 iid($\frac{1}{2}$) vs. period-2 alternation

Proposition A.1. *Let μ be iid Bernoulli($\frac{1}{2}$) and ν the two-point stationary process supported on the alternating sequences. Then $\bar{d}_n(\mu, \nu) = \mathbb{E}[\min(k, n-k)]/n$ with $k \sim \mathrm{Bin}(n, \frac{1}{2})$, and $\bar{d}(\mu, \nu) = \frac{1}{2}$.*

Proof. Q_n is uniform on the two alternating n-blocks $a, \bar{a}$ (bitwise complements); P_n is uniform on $\{0,1\}^n$. For any block b with $k = d_H(b, a)$ mismatches to a, its distance to $\bar{a}$ is $n - k$, so the transport cost from b is at least $\min(k, n-k)/n$; hence $\bar{d}_n \geq \mathbb{E}[\min(k, n-k)]/n$. Assigning each b to its nearer target and splitting complement pairs $\{b, \bar{b}\}$ across the two targets balances the demands ($\frac{1}{2}$ each) exactly, achieving the bound. Monotone convergence of $\mathbb{E}[\min(k, n-k)]/n \to \frac{1}{2}$ follows from $k/n \to \frac{1}{2}$; alternatively, $h(\mu) - h(\nu) = 1$ bit and Prop. 2.2 force $\bar{d} \geq H_{\mathrm{b}}^{-1}(1) = \frac{1}{2}$, while $\bar{d} \leq \frac{1}{2}$ holds because independent coupling of μ with any binary process has cost $\frac{1}{2}$. □

B Implementation details

Integrators. Lorenz-63: fixed-step RK4, $dt = 0.005$, transient 100 time units discarded, seeded random initial conditions. KS: ETDRK4 [13] in rfft space, $N = 64$, $L = 22$, $dt = 0.25$ (0.125 for the speed×2 system), 2/3-rule dealiasing, transient 1000 time units.

IAAFT. Up to 200 iterations, alternating exact rank-remap to the reference amplitude distribution with spectral amplitude replacement; convergence tolerance 10^{-8} on relative spectral error [26].

$\bar{d}$ **estimator.** Exact network-simplex OT (POT [5]); full-distribution regime when both supports ≤ 4096, else fixed budgets of 2000 (multi-seed and KS runs) or 4000 (single-seed convergence study) blocks per side; floors and repeats computed under the identical regime and budget. Block encoding is exact for $|\mathcal{A}|^n < 2^{53}$.

Entropy. Plug-in block entropies with Miller–Madow correction; guard: trust n only if observed support $\leq 0.02\times$ windows. LZ78 cross-check.

Rosenstein λ_1 protocol and its fragility. Delay dim. 3, lag 2, Theiler window 50, fit over steps 1–15 of the mean log-divergence, r^2 of the fit reported; identical protocol for every system. A window study on Lorenz truth (4 seeds, 2×10^5 samples, $\tau = 0.1$) gives $\lambda_1 = 1.195/1.573/1.425/1.683$ nats/t for windows $(1,15)/(1,8)/(2,10)/(1,5)$ (literature 0.906), and speed×2/truth slope ratios of 1.50/1.28/1.34/1.26 against the constructed value 2: early-lag transients inflate short windows, saturation compresses long ones, and no window fixes both. This calibration sensitivity — on the system where the answer is known — is why we treat data-driven Lyapunov estimates as a comparative diagnostic rather than a metric, and it illustrates the practical appeal of certified, protocol-frozen lower bounds like Prop. 2.2.

ESN. Reservoir 400, density 0.02, spectral radius as stated, input scale 0.5, bias $\mathcal{U}(-0.2, 0.2)$, features $[r; r^2]$, ridge 10^{-6}, washout 10^3; training/eval data disjoint and independently seeded.

Compute. Every experiment in this paper runs on a desktop CPU; the complete set reproduces in under two hours single-threaded.